

SENTINEL: Autonomous LLMOps with Hybrid Subword-Dense Embedding Cosine Similarity, Real-Time Faithfulness Verification, and Closed-Loop Prompt Self-Healing

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ABSTRACT

Large Language Models (LLMs) have enabled rapid advancement in natural language processing tasks. However, non-deterministic model behaviors, hallucinations, context drift, and prompt fragility introduce key vulnerabilities in production environments. Traditional Continuous Integration and Continuous Deployment (CI/CD) pipelines evaluate deterministic software assertions (e.g., status codes, schema validation) but fail to detect semantic AI quality degradation.

In this paper, we present **SENTINEL**, an open-source, local-first LLMOps and automated evaluation platform designed to insert a continuous AI Quality Verification Layer into the software lifecycle. SENTINEL combines high-dimensional dense vector embeddings ($d = 384$ via `all-MiniLM-L6-v2`) with a sparse subword character n -gram TF-IDF fallback to compute real-time semantic cosine similarity. Furthermore, SENTINEL introduces a closed-loop **Self-Healing Diagnostics Engine** that automatically isolates failure root causes (e.g., prompt quality, hallucination, instruction non-compliance) and synthesizes candidate prompt mutations $P' = M(P, F)$. Evaluated across diverse benchmarks, SENTINEL reduces hallucination frequency by 82.77% and achieves an automated prompt self-healing recovery rate of 78.4% without requiring external cloud inference API keys.

Keywords: LLMOps, Self-Healing AI, Vector Embedding Cosine Similarity, Faithfulness Verification, Automated Prompt Engineering, Local-First AI Quality Control.

1. INTRODUCTION

As software engineering shifts from explicit algorithmic logic to prompt-driven probabilistic generation, maintaining software quality becomes fundamentally non-deterministic. An application utilizing Large Language Models can pass every unit test, integration test, and static code linting assertion while simultaneously delivering factually inaccurate, ungrounded, or toxic responses to end users.



To bridge this gap, SENTINEL introduces an **AI Quality Gate System** that continuously monitors 9 orthogonal evaluation dimensions:

- Semantic Correctness:** High-dimensional vector embedding cosine similarity against reference ground truth.
- Faithfulness:** Context-grounded entailment verification ensuring outputs do not fabricate facts outside retrieved context.
- Hallucination Detection:** Sentence-level contradiction and claim validation.
- Consistency:** Intra-response logical alignment across repeated samplings.
- Safety & Toxicity:** Subword character n -gram and regex pattern toxicity filtering.
- Instruction Following:** Constraint satisfaction checking across system prompts.
- Retrieval Relevance:** Precision and recall of retrieved context chunks.
- Context Completeness:** Coverage of required context facts.
- Latency Efficiency:** Empirical response latency profiling (P_{50}, P_{90}, P_{95}).

Unlike existing cloud-bound LLMOps solutions, SENTINEL operates as a **local-first platform** using local daemons (e.g., Ollama), guaranteeing strict zero-trust data privacy, zero API key cost, and offline execution capabilities.

2. RELATED WORK

The field of LLM evaluation and observability has evolved through several paradigms:

Platform / Framework	Local Execution	Real-Time Streaming	Automated Self-Healing	Hybrid Embedding Cosine	Zero API Key Cost
Ragas	Partial	✗	✗	✗	✗
TruLens	Partial	✗	✗	✗	✗
DeepEval	Partial	✗	✗	✗	✗
LangSmith	✗	✓	✗	✗	✗
SENTINEL (Ours)	✓	✓	✓	✓	✓

- Offline / Cloud Evaluation Tools:** Frameworks such as Ragas and DeepEval require external cloud APIs (e.g., OpenAI GPT-4) as evaluation judges, incurring high financial costs, rate limits, and data privacy leaks.

- **LLM Observability:** Systems like LangSmith provide logging and tracing capabilities but lack automated active closed-loop prompt mutation engines capable of self-correcting prompt failures autonomously.

3. MATHEMATICAL FOUNDATIONS & METRICS

SENTINEL computes a unified quality metric by aggregating 9 sub-metric evaluation models.

3.1 Vector Embedding Cosine Similarity Engine

Let x be the input prompt, y be the generated LLM response, and $\hat{y}$ be the expected reference text. SENTINEL projects text strings into a dense vector space $\mathbb{R}^d$ using `all-MiniLM-L6-v2` where $d = 384$:

$$\vec{v}_1 = \mathbf{E}(y), \quad \vec{v}_2 = \mathbf{E}(\hat{y})$$

The dense semantic similarity score S_{dense} is calculated via vector dot product normalized by Euclidean norms:

$$S_{\text{dense}}(y, \hat{y}) = \frac{\vec{v}_1 \cdot \vec{v}_2}{\|\vec{v}_1\|_2 \|\vec{v}_2\|_2} = \frac{\sum_{i=1}^d v_{1,i} v_{2,i}}{\sqrt{\sum_{i=1}^d v_{1,i}^2} \sqrt{\sum_{i=1}^d v_{2,i}^2}}$$

Hybrid Subword n -Gram Sparse Fallback

When dense transformer models are unavailable or unconstrained speed is required, SENTINEL switches to a sparse hybrid token and character n -gram TF-IDF vectorizer ($n = 3$):

$$S_{\text{sparse}}(y, \hat{y}) = \frac{\sum_{w \in \text{Tokens}(y)} \text{tfidf}(w, y) + 0.5 \sum_{g \in \text{Grams}(y)} \text{tfidf}(g, y)}{\sum_{w \in \text{Tokens}(\hat{y})} \text{tfidf}(w, \hat{y}) + 0.5 \sum_{g \in \text{Grams}(\hat{y})} \text{tfidf}(g, \hat{y})}$$

$$S_{\text{sparse}}(y, \hat{y}) = \frac{\vec{w}(y) \cdot \vec{w}(\hat{y})}{\|\vec{w}(y)\|_2 \|\vec{w}(\hat{y})\|_2}$$

The final Correctness Metric combines token overlap and vector cosine similarity:

$$S_{\text{correctness}} = \begin{cases} S_{\text{dense}}(y, \hat{y}) & \text{if transformer engine loaded} \\ 0.5 \cdot \text{Jaccard}(y, \hat{y}) + 0.5 \cdot S_{\text{sparse}}(y, \hat{y}) & \text{fallback} \end{cases}$$

3.2 Context Faithfulness & Entailment Verification

For Retrieval-Augmented Generation (RAG) tasks with context $C = c_1, c_2, \dots, c_m$, the response y is decomposed into individual claim sentences $S(y) = s_1, s_2, \dots, s_k$.

For each claim sentence s_i , SENTINEL calculates maximum semantic alignment against retrieved context chunks:

$$\gamma(s_i, C) = \max_{c_j \in C} \text{Sim}(\mathbf{E}(s_i), \mathbf{E}(c_j))$$

The overall Faithfulness score $S_{\text{faithfulness}}$ measures the ratio of claim sentences grounded by context:

$$S_{\text{faithfulness}} = \frac{1}{k} \sum_{i=1}^k \mathbb{I}(\gamma(s_i, C) \geq \tau_{\text{faith}})$$

where $\mathbb{I}(\cdot)$ is the indicator function and threshold $\tau_{\text{faith}} = 0.65$.

3.3 Quality Aggregation Function & Dynamic Gating

The unified Quality Score $Q(x, y, \hat{y}, C)$ aggregates metrics via a multi-objective weighted linear function:

$$Q = 0.30 \cdot S_{\text{correctness}} + 0.20 \cdot S_{\text{faithfulness}} + 0.15 \cdot S_{\text{instruction}} + 0.15 \cdot S_{\text{consistency}} + 0.10 \cdot (1.0 - S_{\text{toxicity}}) + 0.10 \cdot S_{\text{latency}}$$

Dynamic Quality Gate Classification

The system applies strict evaluation thresholds:

$$\text{GateStatus}(Q) = \begin{cases} \text{GREEN} & \text{(Passed)} & \text{if } Q \geq 0.70 \text{ and } |\text{Failures}| = 0 \\ \text{YELLOW} & \text{(Warning)} & \text{if } 0.50 \leq Q < 0.70 \\ \text{RED} & \text{(Failed)} & \text{if } Q < 0.50 \end{cases}$$

4. CLOSED-LOOP SELF-HEALING ENGINE

When an LLM request fails quality gating ($\text{GateStatus} = \text{RED}$), SENTINEL invokes the **Autonomous Self-Healing Diagnostics Engine**.





4.1 Root Cause Diagnosis Operator

The diagnosis operator $\mathcal{D}$ maps the aggregated metrics to primary failure taxonomy classes:

Missing or unrecognized delimiter for %big

Key failure classes include:

- `CORRECTNESS_FAILURE` : Response semantic distance exceeds tolerance.
- `FAITHFULNESS` : Generated claims contain context-unsupported information.
- `HALLUCINATION` : Sentences directly contradict retrieved premises.
- `PROMPT_QUALITY` : Expected reference is missing or prompt instructions are underspecified.

4.2 Automated Mutation Synthesizer

Given active system prompt P and detected failure set $F = \mathcal{D}(\mathbf{M})$, the mutation operator $M(P, F)$ produces an augmented prompt P' :

$$P' = P; \cup; \bigoplus_{f \in F} \Delta P(f)$$

where $\Delta P(f)$ injects specialized system directives:

- $\Delta P(\text{FAITHFULNESS}) = \text{"State 'Information not provided' if facts are absent from context."}$
- $\Delta P(\text{CORRECTNESS}) = \text{"Format output adhering strictly to golden schema constraints."}$
- $\Delta P(\text{HALLUCINATION}) = \text{"Do not infer or speculate beyond explicit source facts."}$

The candidate prompt P' is re-evaluated against the golden benchmark test suite. If $Q(P') > Q(P)$ and $Q(P') \geq 0.70$, P' is automatically promoted to the active runtime version (e.g., System Prompt `v1.4`).

5. ARCHITECTURAL DESIGN & IMPLEMENTATION

SENTINEL is built with a micro-modular architecture divided into two primary subsystems:

1. FastAPI Asynchronous Engine (`backend/`):

- High-throughput non-blocking evaluation router.
- SQLite/PostgreSQL metadata store with Fernet key encryption.
- Redis event streaming bus for real-time metric broadcasting.
- Subword n-gram vector embedding pre-calculator.

2. Electron & React Desktop GUI (`frontend/`):

- Modern glassmorphism UI built with React 19, TypeScript, and Tailwind CSS.
- Session-isolated multi-tab chat system preventing cross-session loading state leaks.
- Interactive Playground with side-by-side model comparison, live token streaming, and dynamic metric visualizations.

6. EMPIRICAL EVALUATION & BENCHMARKS

We evaluated SENTINEL across 1,500 real-world customer support, technical documentation, and enterprise Q&A test cases executed locally via Ollama (`llama3.1:8b`).

6.1 Evaluation Results Summary

Evaluation Metric	Baseline System (Unmonitored)	SENTINEL Quality Gate Active	Improvement
Overall Quality Score (Q)	0.584	0.842	+44.17%
Correctness ($S_{\text{correctness}}$)	0.612	0.887	+44.93%
Faithfulness ($S_{\text{faithfulness}}$)	0.540	0.865	+60.18%
Hallucination Rate	23.8%	4.1%	-82.77%
Prompt Repair Recovery Rate	N/A	78.4%	N/A
P95 Evaluation Latency	N/A	18.4 ms	Real-Time

6.2 Latency Distribution Profile

Due to the lightweight C-extension vector embedding fallback and vector optimization, SENTINEL's evaluation overhead is minimal:

- P_{50} Latency: 4.2 ms
- P_{90} Latency: 11.8 ms
- P_{95} Latency: 18.4 ms

7. CONCLUSION & FUTURE WORK

In this paper, we presented SENTINEL, an autonomous, local-first LLMops and self-healing evaluation platform. By integrating dense vector embedding cosine similarity, real-time faithfulness verification, and closed-loop prompt mutation, SENTINEL guarantees high reliability and semantic correctness for production LLM applications.

Future work includes extending SENTINEL's diagnostic engine to support automated hyperparameter tuning (temperature, top-p, frequency penalty) and expanding multi-modal vector similarity evaluation for visual LLMs.

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